week1_notebook

October 27, 2026

1 HS4002 Week 1 Demo

1.1 Basics of Python

We're working within the **pandas** + **plotnine** framework.

We'll explore a toy dataset about *Love Is Blind* contestants to answer: **to what extent does age, gender, and race affect the likelihood of marriage in LIB?**

[ ]:
```
import pandas as pd
import numpy as np
import plotnine as p9
import warnings
warnings.filterwarnings('ignore')
```

1.2 Loading the Dataset

[ ]:
```
file_location = "lib_toy.csv"
df = pd.read_csv(file_location)
```

2 What does a dataset usually look like?

[ ]:
```
df.head()
```

3 Univariate Analysis

[ ]:
```
df.describe()
```

[ ]:
```
df['Race'].value_counts()
```

3.1 How many seasons of LIB are in this dataset?

[ ]:
```
df['Season'].value_counts()
```

3.2 How many contestants got married?

[ ]:
```
df['Married'].value_counts()
```

3.3 Univariate Plots

Let's make a histogram of the age distribution using `plotnine` — the grammar is identical to ggplot2. Note that column names in `aes()` are quoted strings in Python.

[ ]:
```
(
    p9.ggplot(df) +
    p9.geom_histogram(p9.aes(x='Age'), binwidth=0.5)
)
```

[ ]:
```
fig_fn = "ntu4eva.png"

p = (
    p9.ggplot(df) +
    p9.geom_histogram(p9.aes(x="Age"), binwidth=0.5, fill="red") +
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Count", x="Age", title="Age Distribution of Love is Blind␣
 ↪Contestants")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

[ ]:
```
fig_fn = "ntu4eva.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age"), fill="steelblue") +
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age", title="Age of Love is Blind Contestants")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

4 Does Age Matter?

How does age relate to the likelihood of a contestant getting married?

[ ]:
```
df.groupby('Married')['Age'].mean()
```

[ ]:
```
fig_fn = "age_by_maritalstatus.png"

p = (
```

```
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age", fill="factor(Married)"),␣
 ↪position="identity", alpha=0.5) +
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age",
          title="Age Distribution of Love is Blind Contestants",
          fill="Married")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

5 Let's Throw Race Into the Mix

[ ]:
```
df.groupby(['Married', 'Race'])['Age'].mean()
```

[ ]:
```
fig_fn = "age_marital_race.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age", fill="factor(Married)"),␣
 ↪position="identity", alpha=0.5) +
    p9.facet_wrap("Race") +
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age",
          title="Age Distribution of Love is Blind Contestants",
          fill="Married")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

6 Trying Adding Gender!

[ ]:
```
df = df.assign(Gender=np.where(df['Gender'] == 1, 'Woman', 'Man'))

df.groupby(['Married', 'Gender'])['Age'].mean()
```

[ ]:
```
fig_fn = "age_marital_gender.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age", fill="factor(Married)"),␣
 ↪position="identity", alpha=0.5) +
    p9.facet_wrap("Gender") +
```

```
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age",
         title="Age Distribution of Love is Blind Contestants",
         fill="Married")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```